

Improved Kalman Algorithm for Vertical Height and Velocity Information Fusion of Multi-rotor Unmanned Aerial Vehicle

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Abstract: The paper presents an improved Kalman filter algorithm. It can ensure the precision and stability of the vertical height and velocity information of the multi-rotor unmanned aerial vehicle (UAV), when the statistical characteristics of the system noise and the measurement noise are unknown. The algorithm uses the accelerometer to provide prediction information and adopts the barometric altimeter to provide measurement information. It can estimate the system noise and the measurement noise based on the characteristics of these two sensors. The flight experiment and analysis shows that the obtained vertical height information error is less than 1m, the information update frequency can reach 100Hz, and the good fusion effect of vertical velocity information is obtained. The improved Kalman algorithm can meet the needs of the multi-rotor unmanned aerial vehicle flight control.

Key Words: Kalman filter, Information fusion, Multi-rotor unmanned aerial vehicle, Vertical height and velocity estimation

1 Introduction

Obtaining high-precision, high-update frequency vertical motion information is the key technology to realize the autonomous take-off and landing of the multi-rotor unmanned aerial vehicle (UAV), and is crucial to the stability and safety of low-height flight [1,2].

The accelerometer calculating process of vertical velocity and height has a high update frequency and accuracy in a short time. But it is an integration process and is sensitive to the vibration of the carrier. Over time, the drift errors accumulate and become larger. The barometric altimeter has high accuracy in measuring height information, and there is no integration process. But the accuracy is lower when the barometric altimeter is close to the ground, and it is easily affected by wind disturbances and temperature[3].

Kalman filter is currently recognized as the most suitable algorithm for the information fusion of the multi-sensor navigation, but when the conventional Kalman filtering algorithm is applied, the statistical characteristics of the system noise and the measurement noise need to be known [4,5]. However, in the practical applications of the multi-rotor unmanned aerial vehicle, the statistical characteristics of the system noise and the measurement noise are unknown, which makes the conventional Kalman filter algorithm loses its optimality, the estimation accuracy is greatly reduced, even will cause the filter divergence. This paper presents an improved Kalman filter algorithm. It uses the accelerometer to provide prediction information and adopts the barometric altimeter to provide measurement

information. It takes the noise variance calculated by the accelerometer in the stationary state to estimate the system noise variance, and takes a constant value to estimated measurement noise. Through the improvement of Kalman filter algorithm, a better fusion effect of vertical height and velocity information is obtained, which can meet the needs of the multi-rotor unmanned aerial vehicle flight control.

2 Linear System Modeling

The vertical height and velocity estimation of the multi-rotor unmanned aerial vehicle can be described by linear mathematical model[1].

$$\begin{cases} \dot{h} = v_e \\ \dot{v}_{ez} = a_z^n \end{cases} \quad (1)$$

The state equation of state space model is:

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u \quad (2)$$

where

$\mathbf{x} = [h \quad v_{ez}]^T$ is 2×1-dimensional state vector, h is the vertical height, v_{ez} is the vertical velocity, u is input quantity, $u = a_z^n$, a_z^n is the linear acceleration of the z -axis in the navigation coordinate system,

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

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The research object is the multi-rotor unmanned aerial vehicle used in the civil field. It is generally flying near the ground and has a short navigation distance and time. So the simplification is as follows: Ignore the Coriolis correction, the effect of centripetal force and the rotation of the earth. There is no rotation correction or dynamic correction. Then the navigation equation in the local geographic coordinate system is simplified as:

$$\dot{\mathbf{v}}_e^n = \mathbf{C}_b^n \mathbf{f}_b + \mathbf{g} \quad (3)$$

Where

$\mathbf{v}_e^n = [\mathbf{v}_{ex} \quad \mathbf{v}_{ey} \quad \mathbf{v}_{ez}]^T$ is the ground velocity in the local geographic coordinate system, $\mathbf{f}_b$ is the three-axis output of the accelerometer in the body coordinate system, $\mathbf{C}_b^n$ is strapdown matrix, and $\mathbf{g} = [0 \quad 0 \quad -g]^T$, g is gravitational acceleration. Then,

$$\begin{aligned} u = \dot{\mathbf{v}}_{ez}^n &= a_z^n \\ &= -\sin \theta a_x + \sin \gamma \cos \theta a_y + \cos \gamma \cos \theta a_z - g \end{aligned} \quad (4)$$

Where

θ is the pitch angle, γ is the roll angle, and $\mathbf{a}_b = [a_x \quad a_y \quad a_z]^T$ is the linear acceleration output of the accelerometer in the body coordinate system.

The linear state equation of discrete time is:

$$\mathbf{X}_k = \Phi \mathbf{X}_{k-1} + \mathbf{B}u_{k-1} + \Gamma \mathbf{W}_{k-1} \quad (5)$$

where

$$\Phi = \begin{bmatrix} 1 & T_s \\ 0 & 1 \end{bmatrix}, \mathbf{B} = \begin{bmatrix} \frac{1}{2} T_s^2 \\ T_s \end{bmatrix},$$

$\mathbf{X}_k = [h_k \quad \mathbf{v}_{ek}]^T$ is 2×1 -dimensional state vector, T_s is the information fusion cycle, $\mathbf{W}_k$ is the system noise amount. Its variance statistical characteristics are completely determined by the statistical characteristics of the model input, that is, by the statistical characteristics of the accelerometer. Taking the system noise matrix is $\Gamma = \mathbf{B}$.

The measurement equation of discrete time state space model is:

$$\mathbf{Z}_k = \mathbf{H}\mathbf{X}_k + \mathbf{V}_k \quad (6)$$

Where

$\mathbf{Z}_k$ is the measurement amount, which is provided by the height information output of the barometric altimeter. $\mathbf{V}_k$ is the measurement noise amount. Considering that the noise error of the barometric altimeter cannot be simply considered as white noise, its statistical characteristics are unknown.

$\mathbf{H} = [1 \quad 0]$ is 1×2 -dimensional constant coefficient measurement matrix.

To sum up, the discrete time linear state space model of the vertical height and velocity information fusion system for the multi-rotor unmanned aerial vehicle is as follows:

$$\begin{cases} \mathbf{X}_k = \Phi \mathbf{X}_{k-1} + \mathbf{B}u_{k-1} + \Gamma \mathbf{W}_{k-1} \\ \mathbf{Z}_k = \mathbf{H}\mathbf{X}_k + \mathbf{V}_k \end{cases} \quad (7)$$

that is

$$\begin{cases} \begin{bmatrix} h_k \\ \mathbf{v}_{ek} \end{bmatrix} = \begin{bmatrix} 1 & T_s \\ 0 & 1 \end{bmatrix} \begin{bmatrix} h_{k-1} \\ \mathbf{v}_{ek-1} \end{bmatrix} + \begin{bmatrix} \frac{1}{2} T_s^2 \\ T_s \end{bmatrix} (a_{zk-1}^n + \mathbf{W}_{k-1}) \\ \mathbf{Z}_k = [1 \quad 0] \begin{bmatrix} h_k \\ \mathbf{v}_{ek} \end{bmatrix} + \mathbf{V}_k \end{cases} \quad (8)$$

3 Design of Improved Kalman Filter Algorithm

For the improved Kalman filter algorithm, in a information fusion cycle, the complete estimation steps are:

- State single-step predicting

$$\hat{\mathbf{X}}_{k/k-1} = \Phi \hat{\mathbf{X}}_{k-1} + \mathbf{B}u_{k-1} \quad (9)$$

- Innovation sequence updating

$$\mathbf{v}_k = \mathbf{Z}_k - \mathbf{H} \hat{\mathbf{X}}_{k/k-1} \quad (10)$$

- State one-step prediction mean square error matrix update

$$\mathbf{P}_{k/k-1} = \Phi \mathbf{P}_{k-1} \Phi^T + \Gamma \hat{\mathbf{Q}}_{k-1} \Gamma^T \quad (11)$$

- Measurement noise estimating

$$\hat{\mathbf{R}}_k = (\mathbf{I} - \beta_k) \hat{\mathbf{R}}_{k-1} + \beta_k [(\mathbf{I} - \mathbf{H}\mathbf{K}_{k-1}) \mathbf{v}_k \mathbf{v}_k^T (\mathbf{I} - \mathbf{H}\mathbf{K}_{k-1})^T + \mathbf{H}\mathbf{P}_{k-1} \mathbf{H}^T] \quad (12)$$

- Filter gain updating

$$\mathbf{K}_k = \frac{\mathbf{P}_{k/k-1} \mathbf{H}^T}{\mathbf{H}\mathbf{P}_{k/k-1} \mathbf{H}^T + \hat{\mathbf{R}}_k} \quad (13)$$

- State estimating

$$\hat{X}_k = \hat{X}_{k/k-1} + K_k v_k \quad (14)$$

- State estimation mean square error matrix update

$$P_k = (I - K_k H) P_{k/k-1} (I - K_k H)^T + K_k R_k K_k^T \quad (15)$$

- System noise estimating

$$\hat{Q}_k = \text{Acceleration variances calculated by accelerometer} \quad (16)$$

The algorithm only needs given initial value X_0 , P_0 , Q_0 , just it can get the vertical height and velocity state estimation $\hat{X}_k$ at time k by the recursive calculation [6].

The statistical characteristics of the system noise in the model are completely determined by the accelerometer. Obtaining the vertical height and velocity information, through an accelerometer, is an integration process. It is not suitable to use an adaptive algorithm to estimate the system noise $\hat{Q}_k$ in real time. The variance calculated by the accelerometer in a stationary state can be used as the system noise. The method is to sample the accelerometer output information for a period of time before each takeoff of the unmanned aerial vehicle, and the output variance calculated is used as the statistical characteristics of the system noise of this flight. At this time, the unmanned aerial vehicle is stationary relative to the ground and the engine is not started. The measurement noise $\hat{R}_k$ is determined by the barometric altimeter, and which needs to be estimated adaptively according to Formula (12).

4 Experiment and analysis

4.1 Experiment Conditions

The experimental data are acquired from the multi-rotor unmanned aerial vehicle carrying the ADIS16488 inertial measurement unit (IMU). The inertial measurement unit has a three-axis accelerometer and a barometric altimeter. The initial sensitivity of the accelerometer is (1.221×10^{-8}) g/LSB, and the initial sensitivity of the barometric altimeter is (6.1×10^{-7}) mbar/LSB.

- The accelerometer output information is updated once every $T_A \approx 0.01s$. and the barometric altimeter is updated once every $T_B = 0.02s$. Every $T_K = 0.02s$, The improved Kalman filter algorithm is used to estimate vertical height and velocity information.
- Using the information calculated by the accelerometer is to interpolate between each T_K cycle, the frequency of information fusion can reach $f_h = 100Hz$.
- Evaluation method for the effectiveness of the algorithm: In flight experiments, the barometric altimeter has high accuracy when it is far from the ground. Therefore, when the multi-rotor unmanned aerial vehicle is flying stably in the air, the barometric altimeter output can be used as a standard reference curve.

4.2 Results and Analysis

According to the improved Kalman filter algorithm, the statistical characteristics of the measured noise is adaptively estimated in real time. The experimental curve is shown in Fig. 1. The three curves in the figure are the height curve of barometric altimeter, the vertical height information fusion curve and the vertical velocity information fusion curve.

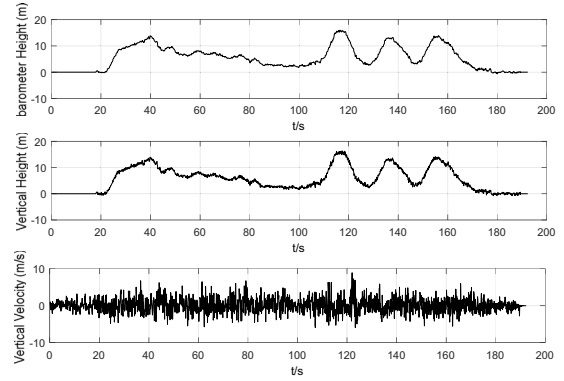


Fig. 1: The vertical height and velocity curves of the improved Kalman filter algorithm($\hat{R}_k$ is estimated adaptively.)

The experimental results show that the root mean square error of the fusion height is less than 1m, which basically meets the requirements, but the curve is not smooth. In addition, the fusion effect of vertical velocity is very poor and cannot be directly used for flight control. Therefore, the designed algorithm needs to be modified.

Through analysis, it can be known that the body vibration of the multi-rotor unmanned aerial vehicle is the main source of disturbance for navigation information measurement [7]. The main frequency of body vibration is about 35Hz, and the output update frequency of the barometric altimeter is 50Hz, their frequency are very close. So there is a large error in the measurement noise estimation value calculated by Formula (12), and the statistical characteristics of measurement noise cannot be truly expressed. Therefore, the measurement noise estimation is determined to be constant through multiple flight experiments,

$$\hat{R}_k = d \quad (17)$$

Φ , H , and Γ in Formula (7) are constant matrices, so the Formula (11) is modified as

$$P_{k/k-1} = \Phi P_0 \Phi^T + \Gamma \hat{Q} \Gamma^T = P \quad (18)$$

the Formula (13) is modified as

$$K = \frac{PH^T}{HPH^T + R} \quad (19)$$

According to the modified algorithm, the experimental curve is shown in Fig. 2.

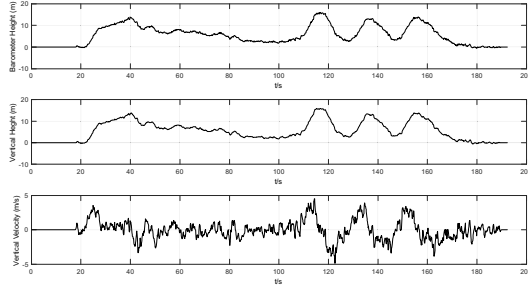


Fig. 2: The vertical height and velocity curves of the modified algorithm($\hat{R}_k$ is constant)

The fusion effect of the modified algorithm is better, which can meet the needs of the multi-rotor unmanned aerial vehicle flight control. However, the vertical velocity fusion accuracy comparison cannot be performed at present, and further analysis and research are needed.

5 Conclusion

● The improved Kalman filter algorithm is simple and effective, which can correctly judge the sensor characteristics of the accelerometer and barometric altimeter, and ensure the accuracy of the estimated value.

● The flight experiment was performed on a multi-rotor unmanned aerial vehicle carrying an inertial

measurement unit, and the vertical height error was less than 1m. By observing the experimental curves, the vertical velocity information fusion effect is good, but it is currently impossible to make a quantitative analysis of the accuracy.

- The improved Kalman filter algorithm can make the vertical height and velocity information update frequency reach 100Hz, which is more in line with the high-frequency dynamic changes of the multi-rotor unmanned aerial vehicle.

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